PROBLEM 37

Particle on a Spinning Elliptic Cone

Original statement · Problems.pdf, p. 25

Rotating Constraint Surface

A heavy point particle P with mass m moves without friction on a cone of elliptic cross-section as shown in the figure. We define an auxiliary reference frame $S_1 : \{O; \mathcal{B}_1\}$ attached to the cone. The cone is defined in term [sic] of two parameters ζ, β as

$$\text{Elliptic cone:} \quad \begin{cases} x_1 = \zeta \cos\beta \\ y_1 = 2\zeta \sin\beta \\ z_1 = \zeta \end{cases} \qquad (5)$$

The cone spins about its Oz_1 axis at a constant angular rate Ω with respect to the inertial reference frame $S_0 : \{O; \mathcal{B}_0\}$. At time $t = 0$, the reference frame S_1 coincides with S_0, the particle P is located at $\mathbf{r}^P_{1,0} = 2\mathbf{j}_1 + \mathbf{k}_1$, and has a velocity $\mathbf{v}^P_{1,0} = \mathbf{i}_1$.

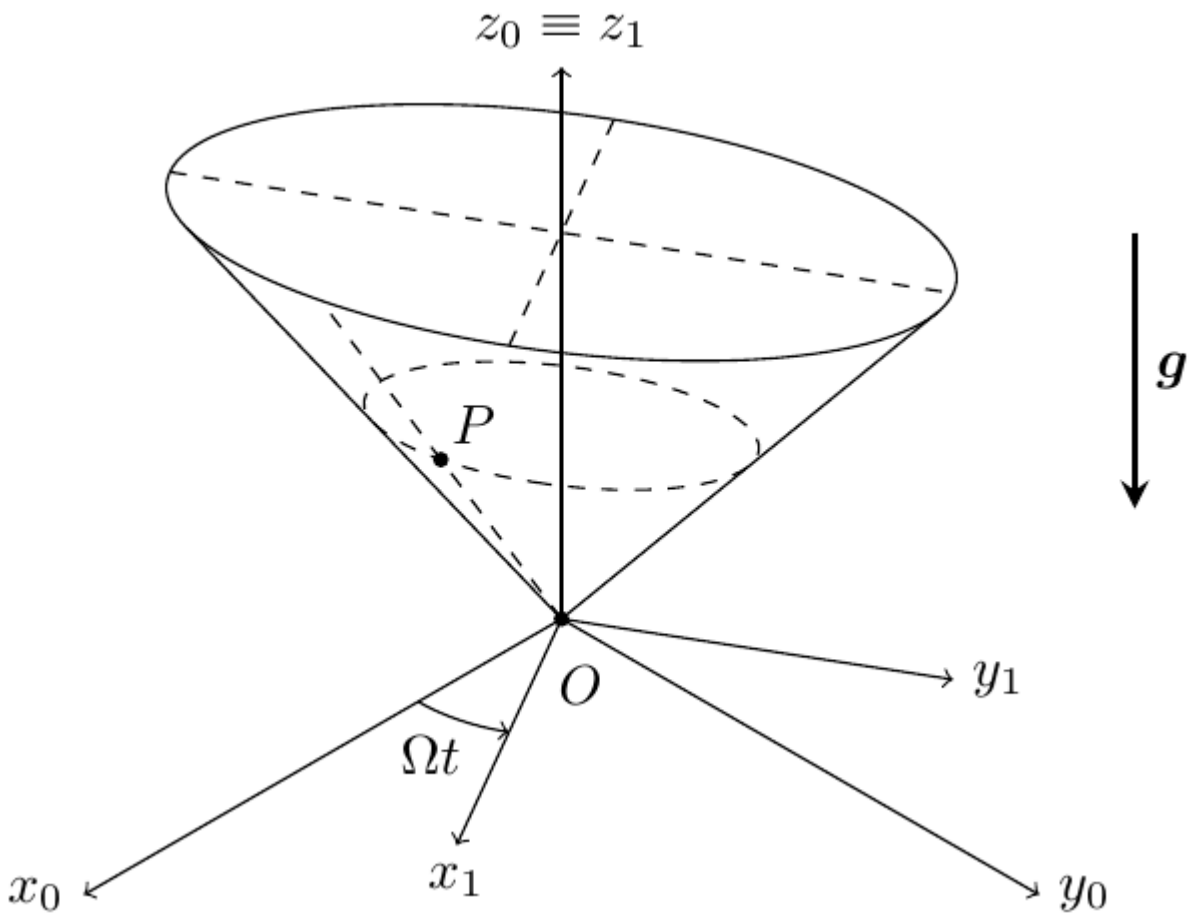


Figure: Elliptic cone with vertex O and vertical axis z0 = z1, frame S1 rotated by the angle Omega t about z0, particle P on the cone, gravity g downward (Problems.pdf, p. 25).

(a) How many degrees of freedom does the particle motion have?

(b) Obtain the general expression for the position, velocity, and acceleration of P **in reference frame** S_1 as a function of ζ, β and their derivatives. Remember to define any unit vectors that you use in terms of the vectors of $\mathcal{B}_0$.

(c) Identify and express mathematically all the forces acting on P as seen from S_1.

(d) Obtain two second-order differential equations for the evolution of ζ, β in time, which do not depend on any other unknowns in the problem.

(e) Find, justify, and write down one conservation law in this problem, including the value of any constants there might be.

(f) Find the equilibrium positions for P in S_1 for $\zeta > 0$ as a function of the constants of the problem.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** a heavy point particle P of mass m slides on the inner surface of an elliptic cone with vertex O and vertical axis $Oz_1 \equiv Oz_0$. The cone spins about that axis with constant angular rate Ω relative to the inertial frame S_0.
- **Hypotheses:**
  1. $S_0 = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$ is inertial; gravity is $\mathbf{g} = -g\,\mathbf{k}_0$.
  2. The contact is smooth (no friction) and bilateral: the reaction of the cone is normal to its surface.
  3. Ω is constant and the vertex O is fixed, so O is a common origin of S_0 and S_1.
  4. The data $\mathbf{r}^P_{1,0} = 2\mathbf{j}_1 + \mathbf{k}_1$ and $\mathbf{v}^P_{1,0} = \mathbf{i}_1$ are read in SI units (m and m/s).
- **Why a moving frame:** in S_0 the surface of the cone moves, so the constraint depends explicitly on time. In the frame S_1 attached to the cone the surface is fixed and is described by the two parameters (ζ, β) of Eq. (5). The price is that S_1 is not inertial, so Newton's law in S_1 needs the inertia forces of Notes Eqs. 4.62–4.63 (Slides 27–28); they are known functions of the relative position and velocity.
- **Fixed (inertial) frame:** $S_0 = \{O; \mathcal{B}_0\}$, $\mathcal{B}_0 = \{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, with $\mathbf{k}_0$ vertical and upward.
- **Cone frame:** $S_1 = \{O; \mathcal{B}_1\}$, $\mathcal{B}_1 = \{\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, attached to the cone, with $\mathbf{k}_1 = \mathbf{k}_0$ along its axis and $\mathbf{i}_1, \mathbf{j}_1$ along the two axes of its elliptic sections. It turns the angle Ωt about $\mathbf{k}_0$ (the figure of the statement measures Ωt from x_0 to x_1) and coincides with S_0 at $t = 0$.
- **Surface basis:** the two tangent vectors $\mathbf{e}_\zeta, \mathbf{e}_\beta$ of the cone at P and its unit normal $\mathbf{n}$, deduced in the next section.
- **Coordinates:** (ζ, β), the parameters of Eq. (5): ζ is the height of P above the vertex and β the angular parameter of the elliptic section through P.

Symbol	Meaning	SI unit
m	mass of P	kg
g	gravitational acceleration	m/s^2
Ω	constant spin rate of the cone	rad/s
ζ	height of P above the vertex ($z_1 = \zeta$)	m
β	angular parameter of the elliptic section	rad
N	magnitude of the normal reaction	N

(a) Degrees of freedom

> **Pedagogical Justification:** a free particle has 3 coordinates; each independent holonomic constraint removes one.

Eliminating the parameters of Eq. (5): $\cos\beta = x_1/\zeta$, $\sin\beta = y_1/(2\zeta)$ and $\zeta = z_1$, so $\cos^2\beta + \sin^2\beta = 1$ gives the single constraint equation of the surface, fixed in S_1:

$$\frac{x_1^2}{z_1^2} + \frac{y_1^2}{4z_1^2} = 1 \iff x_1^2 + \frac{y_1^2}{4} - z_1^2 = 0$$

Hence $\mathrm{DOF} = 3 - 1 = 2$, and (ζ, β) are suitable generalized coordinates. Each horizontal section $z_1 = \zeta$ is an ellipse with semi-axes ζ (along $\mathbf{i}_1$) and 2ζ (along $\mathbf{j}_1$).

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

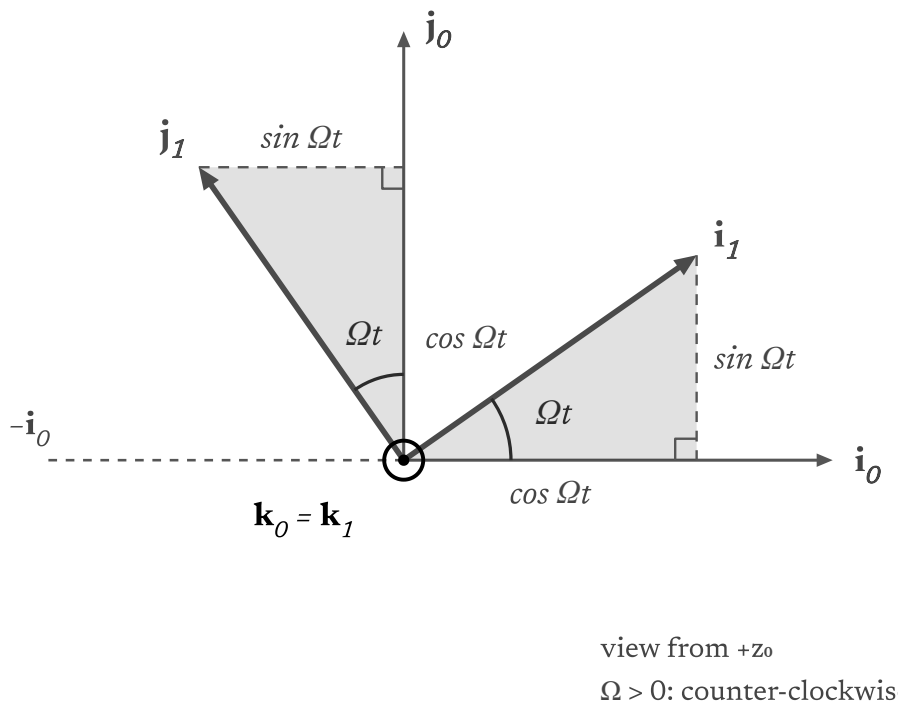


Figure 1. View from $+z_0$, with $\mathbf{k}_0 = \mathbf{k}_1$ out of the paper ($\odot$), so $\mathbf{j}_0 = \mathbf{k}_0 \times \mathbf{i}_0$ points up. $\mathbf{i}_1, \mathbf{j}_1$ are $\mathbf{i}_0, \mathbf{j}_0$ turned counter-clockwise by Ωt: $\mathbf{i}_1$ has legs $\cos \Omega t$ along $\mathbf{i}_0$ and $\sin \Omega t$ along $\mathbf{j}_0$; $\mathbf{j}_1$ has legs $\cos \Omega t$ along $\mathbf{j}_0$ and $\sin \Omega t$ along $-\mathbf{i}_0$, hence its minus sign.

The cone spins about $\mathbf{k}_0$, so we look down the axis from $+z_0$: $\mathbf{k}_0 = \mathbf{k}_1$ points out of the paper, towards the reader ($\odot$). Take $\mathbf{i}_0$ to the right; then $\mathbf{j}_0 = \mathbf{k}_0 \times \mathbf{i}_0$ points up the page (right-hand rule: fingers from $\mathbf{k}_0$, towards the reader, to $\mathbf{i}_0$, right; the thumb points up). With $\Omega > 0$ the cone turns counter-clockwise in this view (thumb along $\mathbf{k}_0$), and at time t it has turned Ωt from $\mathbf{i}_0$:

- $\mathbf{i}_1$ is the hypotenuse (length 1) of the shaded right triangle on the right: the leg adjacent to Ωt lies along $\mathbf{i}_0$ and measures $\cos \Omega t$; the opposite leg lies along $+\mathbf{j}_0$ and measures $\sin \Omega t$.
- $\mathbf{j}_1$ starts on $\mathbf{j}_0$ (up) and turns the same angle counter-clockwise, so its tip leans to the left, towards $-\mathbf{i}_0$: its leg along $\mathbf{j}_0$ (adjacent) is $\cos \Omega t$ and its leg along $-\mathbf{i}_0$ (opposite) is $\sin \Omega t$. **That is where the minus sign of $\mathbf{j}_1$ comes from.**
- $\mathbf{k}_1 = \mathbf{k}_0$, the axis of the rotation.

$$\mathbf{i}_1 = \cos\Omega t\,\mathbf{i}_0 + \sin\Omega t\,\mathbf{j}_0, \qquad \mathbf{j}_1 = -\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0, \qquad \mathbf{k}_1 = \mathbf{k}_0$$

The columns of the change-of-basis matrix are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\Omega t & -\sin\Omega t & 0 \\ \sin\Omega t & \cos\Omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos\Omega t\cos\Omega t - (-\sin\Omega t)\sin\Omega t = \cos^2\Omega t + \sin^2\Omega t = 1$$

In the product $[{}_0R_1][{}_0R_1]^T$ the diagonal entries are $\cos^2\Omega t + \sin^2\Omega t = 1$ (first and second rows) and 1 (third row), and the off-diagonal ones are $\cos\Omega t\sin\Omega t - \sin\Omega t\cos\Omega t = 0$ (first row times second) and 0 (any row times the third), so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1: the origin is fixed, $\mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$. For a simple rotation about the fixed vector $\mathbf{k}_0$ (Slide 16), $\boldsymbol{\omega}_{10} = \Omega\,\mathbf{k}_1$. Check with the chain rule: $\left.\dfrac{d\mathbf{i}_1}{dt}\right|_0 = \dfrac{d\cos\Omega t}{d(\Omega t)}\,\Omega\,\mathbf{i}_0 + \dfrac{d\sin\Omega t}{d(\Omega t)}\,\Omega\,\mathbf{j}_0 = \Omega\,(-\sin\Omega t\,\mathbf{i}_0 + \cos\Omega t\,\mathbf{j}_0) = \Omega\,\mathbf{j}_1 = \Omega\,\mathbf{k}_1 \times \mathbf{i}_1$. Since Ω and $\mathbf{k}_1 = \mathbf{k}_0$ are constant, $\boldsymbol{\alpha}_{10} = \left.\dfrac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \mathbf{0}$.

The elliptic section through P

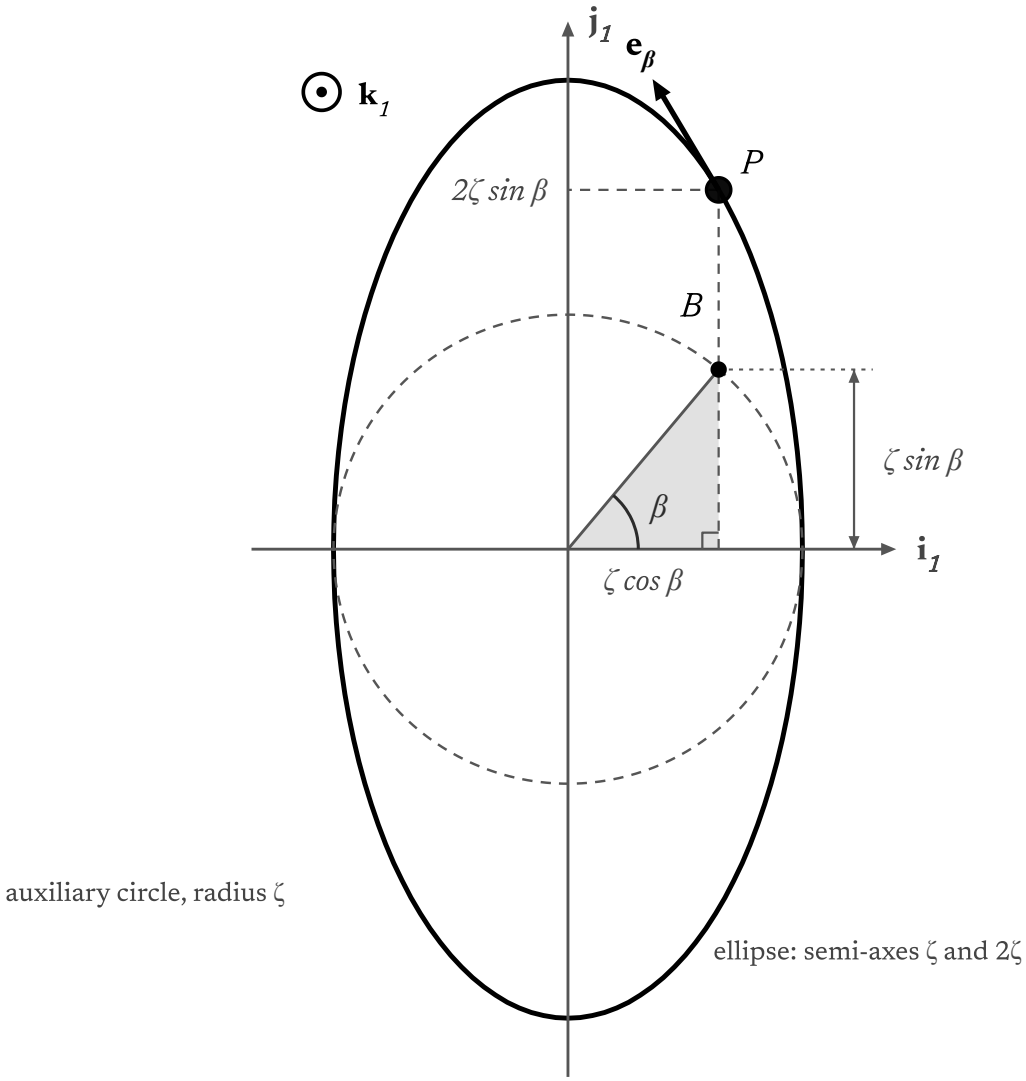


Figure 2. Section $z_1 = \zeta$ seen from above ($\mathbf{k}_1$ out of the paper). β is the polar angle of the point B of the dashed auxiliary circle of radius ζ; in the shaded triangle the legs are $\zeta\cos\beta$ and $\zeta\sin\beta$. The projection of P has the same $x_1 = \zeta\cos\beta$ and twice the height, $y_1 = 2\zeta\sin\beta$; $\mathbf{e}_\beta$ (orange) is tangent to the ellipse.

O is the origin of S_1, so $\mathbf{r}_1^P = \overrightarrow{OP}$ (Notes Eq. 3.16 with $\mathbf{r}_0^O = \mathbf{0}$). We build it in two steps: up the axis to the height of P, then across the horizontal section. The section $z_1 = \zeta$, seen from above, is the ellipse of semi-axes ζ along $\mathbf{i}_1$ and 2ζ along $\mathbf{j}_1$.

What β is. β is not the polar angle of P. Draw the auxiliary circle of radius ζ (dashed) and the point B of that circle at the polar angle β. In the shaded right triangle with vertex O and hypotenuse $OB = \zeta$, the leg adjacent to β lies along $\mathbf{i}_1$ and measures $\zeta\cos\beta$, and the opposite leg measures $\zeta\sin\beta$. The point of the ellipse has the same $x_1 = \zeta\cos\beta$ and twice the height, $y_1 = 2\zeta\sin\beta$: the ellipse is the circle stretched by a factor 2 along $\mathbf{j}_1$. So β is the polar angle of B, not of P; the polar angle α of the horizontal projection of P satisfies

$$\tan\alpha = \frac{y_1}{x_1} = \frac{2\zeta\sin\beta}{\zeta\cos\beta} = 2\tan\beta$$

and $\alpha = \beta$ only at $\beta = 0, \pi/2, \pi, 3\pi/2$, on the axes of the ellipse. Adding the vertical part $\zeta\,\mathbf{k}_1$:

$$\mathbf{r}_1^P = \zeta\cos\beta\,\mathbf{i}_1 + 2\zeta\sin\beta\,\mathbf{j}_1 + \zeta\,\mathbf{k}_1$$

Check with the initial datum: $\zeta = 1, \beta = \pi/2$ give $2\mathbf{j}_1 + \mathbf{k}_1 = \mathbf{r}_{1,0}^P$.

Tangent vectors and normal of the cone. Two tangent vectors at P are the partial derivatives of $\mathbf{r}_1^P$ with respect to each parameter (they are not unit vectors). With $\frac{\partial}{\partial\beta}\cos\beta = -\sin\beta$ and $\frac{\partial}{\partial\beta}\sin\beta = \cos\beta$:

$$\mathbf{e}_\zeta \equiv \frac{\partial\mathbf{r}_1^P}{\partial\zeta} = \cos\beta\,\mathbf{i}_1 + 2\sin\beta\,\mathbf{j}_1 + \mathbf{k}_1, \qquad \frac{\partial\mathbf{r}_1^P}{\partial\beta} = \zeta\,\mathbf{e}_\beta, \quad \mathbf{e}_\beta \equiv -\sin\beta\,\mathbf{i}_1 + 2\cos\beta\,\mathbf{j}_1$$

$\mathbf{e}_\zeta$ runs along the generator through P (the straight line from the vertex) and $\mathbf{e}_\beta$ is tangent to the ellipse, drawn at P in the figure. A normal vector is their cross product, expanded component by component:

$$\mathbf{e}_\zeta \times \mathbf{e}_\beta = \begin{vmatrix} \mathbf{i}_1 & \mathbf{j}_1 & \mathbf{k}_1 \\ \cos\beta & 2\sin\beta & 1 \\ -\sin\beta & 2\cos\beta & 0 \end{vmatrix} = (0 - 2\cos\beta)\,\mathbf{i}_1 - (0 + \sin\beta)\,\mathbf{j}_1 + \left(2\cos^2\beta + 2\sin^2\beta\right)\mathbf{k}_1 = -2\cos\beta\,\mathbf{i}_1 - \sin\beta\,\mathbf{j}_1 + 2\,\mathbf{k}_1$$

Its squared norm is $4\cos^2\beta + \sin^2\beta + 4 = 5 + 3\cos^2\beta$, so the unit normal (pointing into the cone, upward) is

$$\mathbf{n} = \frac{-2\cos\beta\,\mathbf{i}_1 - \sin\beta\,\mathbf{j}_1 + 2\,\mathbf{k}_1}{\sqrt{5 + 3\cos^2\beta}}$$

(Check: the gradient of $x_1^2 + y_1^2/4 - z_1^2$ is $(2x_1, y_1/2, -2z_1) = \zeta(2\cos\beta, \sin\beta, -2)$, parallel to $\mathbf{n}$.)

The two principal vertical sections

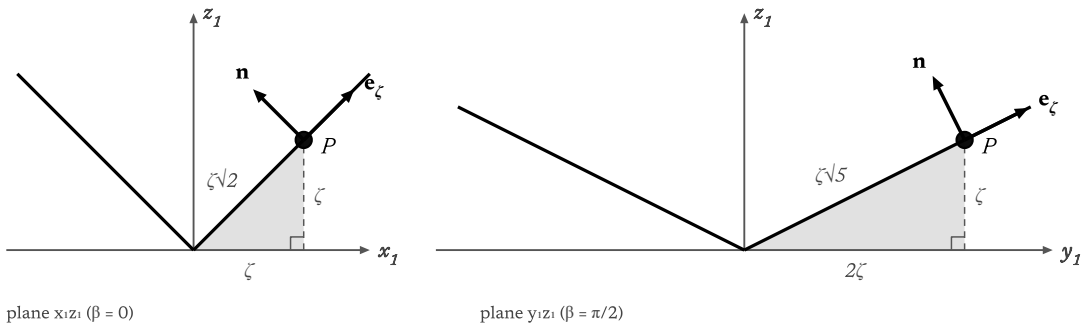


Figure 3. Principal vertical sections. Left, plane x_1z_1 ($\beta = 0$): legs ζ and ζ, hypotenuse $\zeta\sqrt{2}$, $\mathbf{e}_\zeta = \mathbf{i}_1 + \mathbf{k}_1$. Right, plane y_1z_1 ($\beta = \pi/2$): legs 2ζ and ζ, hypotenuse $\zeta\sqrt{5}$, $\mathbf{e}_\zeta = 2\mathbf{j}_1 + \mathbf{k}_1$. The normal $\mathbf{n}$ is perpendicular to the generator in each plane.

In the vertical plane x_1z_1 ($\beta = 0$ or π) the generators are the lines $z_1 = |x_1|$: in the shaded right triangle the horizontal leg is ζ and the vertical leg is ζ, so the hypotenuse is $\zeta\sqrt{2}$ and the generator makes 45° with the horizontal. In the plane y_1z_1 ($\beta = \pm\pi/2$) the generators are $z_1 = |y_1|/2$: horizontal leg 2ζ, vertical leg ζ, hypotenuse $\sqrt{4\zeta^2 + \zeta^2} = \zeta\sqrt{5}$. At $\beta = 0$, $\mathbf{e}_\zeta = \mathbf{i}_1 + \mathbf{k}_1$ and $\mathbf{n} = (-\mathbf{i}_1 + \mathbf{k}_1)/\sqrt{2}$; at $\beta = \pi/2$, $\mathbf{e}_\zeta = 2\mathbf{j}_1 + \mathbf{k}_1$ and $\mathbf{n} = (-\mathbf{j}_1 + 2\mathbf{k}_1)/\sqrt{5}$, perpendicular to the generator in each drawing. These sections are where the equilibria of part (f) lie.

Kinematics

(b) Position, velocity and acceleration in S_1

> **Pedagogical Justification:** the relative derivatives $\frac{d}{dt}\big|_1$ treat $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ as constants, so only the components of $\mathbf{r}_1^P$ are differentiated (product and chain rules).

$$\mathbf{r}_1^P = \zeta\cos\beta\,\mathbf{i}_1 + 2\zeta\sin\beta\,\mathbf{j}_1 + \zeta\,\mathbf{k}_1$$

Component by component, with $\frac{d}{dt}\cos\beta = \frac{d\cos\beta}{d\beta}\dot\beta = -\dot\beta\sin\beta$ and $\frac{d}{dt}\sin\beta = \dot\beta\cos\beta$:

$$\frac{d}{dt}(\zeta\cos\beta) = \dot\zeta\cos\beta - \zeta\dot\beta\sin\beta, \qquad \frac{d}{dt}(2\zeta\sin\beta) = 2\dot\zeta\sin\beta + 2\zeta\dot\beta\cos\beta, \qquad \frac{d}{dt}\zeta = \dot\zeta$$

$$\mathbf{v}_1^P = \Big(\dot\zeta\cos\beta - \zeta\dot\beta\sin\beta\Big)\,\mathbf{i}_1 + 2\Big(\dot\zeta\sin\beta + \zeta\dot\beta\cos\beta\Big)\,\mathbf{j}_1 + \dot\zeta\,\mathbf{k}_1$$

Equivalently $\mathbf{v}_1^P = \dot\zeta\,\mathbf{e}_\zeta + \zeta\dot\beta\,\mathbf{e}_\beta$: the velocity is tangent to the cone.

Differentiating again. For the $\mathbf{i}_1$ component, $\frac{d}{dt}(\dot\zeta\cos\beta) = \ddot\zeta\cos\beta - \dot\zeta\dot\beta\sin\beta$ and $\frac{d}{dt}(\zeta\dot\beta\sin\beta) = \dot\zeta\dot\beta\sin\beta + \zeta\ddot\beta\sin\beta + \zeta\dot\beta^2\cos\beta$; for the $\mathbf{j}_1$ component, $\frac{d}{dt}(\dot\zeta\sin\beta) = \ddot\zeta\sin\beta + \dot\zeta\dot\beta\cos\beta$ and $\frac{d}{dt}(\zeta\dot\beta\cos\beta) = \dot\zeta\dot\beta\cos\beta + \zeta\ddot\beta\cos\beta - \zeta\dot\beta^2\sin\beta$. Collecting terms:

$$\mathbf{a}_1^P = \Big[\Big(\ddot\zeta - \zeta\dot\beta^2\Big)\cos\beta - \Big(2\dot\zeta\dot\beta + \zeta\ddot\beta\Big)\sin\beta\Big]\,\mathbf{i}_1 + 2\Big[\Big(\ddot\zeta - \zeta\dot\beta^2\Big)\sin\beta + \Big(2\dot\zeta\dot\beta + \zeta\ddot\beta\Big)\cos\beta\Big]\,\mathbf{j}_1 + \ddot\zeta\,\mathbf{k}_1$$

The unit vectors $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ are defined in terms of $\mathcal{B}_0$ in the change-of-basis section. For later use we write $\mathbf{r}_1^P = x_1\mathbf{i}_1 + y_1\mathbf{j}_1 + z_1\mathbf{k}_1$ and $\mathbf{v}_1^P = \dot x_1\mathbf{i}_1 + \dot y_1\mathbf{j}_1 + \dot z_1\mathbf{k}_1$ with the components above.

Forces and Equations of Motion

(c) Forces acting on P as seen from S_1

> **Pedagogical Justification:** in the non-inertial frame S_1 Newton's law reads $m\mathbf{a}_1^P = \mathbf{F} + \mathbf{F}_i$ (Notes Eq. 4.63), where $\mathbf{F}$ are the actual forces and $\mathbf{F}_i$ the inertia forces of Notes Eq. 4.62 (Slide 28).

1. Weight: $\mathbf{W} = -mg\,\mathbf{k}_0 = -mg\,\mathbf{k}_1$.

2. Normal reaction. The contact is smooth, so the reaction is along the unit normal of the change-of-basis section:

$$\mathbf{N} = N\,\mathbf{n}, \qquad \mathbf{n} = \frac{-2\cos\beta\,\mathbf{i}_1 - \sin\beta\,\mathbf{j}_1 + 2\,\mathbf{k}_1}{\sqrt{5 + 3\cos^2\beta}}$$

with N an unknown scalar.

3. Inertia forces (Notes Eq. 4.62). Since $\mathbf{a}_0^O = \mathbf{0}$ and $\boldsymbol{\alpha}_{10} = \mathbf{0}$, the force due to the acceleration of the origin and the Euler force vanish: $\mathbf{F}_A = -m\mathbf{a}_0^O = \mathbf{0}$, $\mathbf{F}_E = -m\boldsymbol{\alpha}_{10} \times \mathbf{r}_1^P = \mathbf{0}$.

Centrifugal force. With $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$, $\mathbf{k}_1 \times \mathbf{k}_1 = \mathbf{0}$:

$$\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P = \Omega\,\mathbf{k}_1 \times (x_1\mathbf{i}_1 + y_1\mathbf{j}_1 + z_1\mathbf{k}_1) = \Omega\,(x_1\mathbf{j}_1 - y_1\mathbf{i}_1)$$

$$\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) = \Omega\,\mathbf{k}_1 \times \Omega\,(x_1\mathbf{j}_1 - y_1\mathbf{i}_1) = -\Omega^2\,(x_1\mathbf{i}_1 + y_1\mathbf{j}_1)$$

$$\mathbf{F}_C = -m\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^P\right) = m\Omega^2\,(x_1\mathbf{i}_1 + y_1\mathbf{j}_1) = m\Omega^2\zeta\,(\cos\beta\,\mathbf{i}_1 + 2\sin\beta\,\mathbf{j}_1)$$

which is $m\Omega^2\mathbf{r}_\perp$, radially outward from the axis (Slide 30).

Coriolis force.

$$\mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10} \times \mathbf{v}_1^P = -2m\Omega\,\mathbf{k}_1 \times (\dot{x}_1\mathbf{i}_1 + \dot{y}_1\mathbf{j}_1 + \dot{z}_1\mathbf{k}_1) = 2m\Omega\,(\dot{y}_1\,\mathbf{i}_1 - \dot{x}_1\,\mathbf{j}_1)$$

$$\mathbf{F}_{COR} = 4m\Omega\left(\dot{\zeta}\sin\beta + \zeta\dot{\beta}\cos\beta\right)\mathbf{i}_1 - 2m\Omega\left(\dot{\zeta}\cos\beta - \zeta\dot{\beta}\sin\beta\right)\mathbf{j}_1$$

Equation of motion in S_1 (Notes Eq. 4.63):

$$m\mathbf{a}_1^P = -mg\,\mathbf{k}_1 + N\,\mathbf{n} + \mathbf{F}_C + \mathbf{F}_{COR}$$

(d) Two equations of motion free of the reaction

> **Pedagogical Justification:** the only unknown besides ζ and β is N, and $\mathbf{N}$ is perpendicular to the surface. Projecting the vector equation on the two tangent vectors $\mathbf{e}_\zeta$ and $\mathbf{e}_\beta$ eliminates it: $(-2\cos\beta)\cos\beta + (-\sin\beta)(2\sin\beta) + 2 = 0$ and $(-2\cos\beta)(-\sin\beta) + (-\sin\beta)(2\cos\beta) = 0$.

Projection on $\mathbf{e}_\zeta = (\cos\beta, 2\sin\beta, 1)$. Left-hand side:

$$\mathbf{a}_1^P \cdot \mathbf{e}_\zeta = \left(\ddot{\zeta} - \zeta\dot{\beta}^2\right)\left(\cos^2\beta + 4\sin^2\beta\right) + \left(2\dot{\zeta}\dot{\beta} + \zeta\ddot{\beta}\right)(-\sin\beta\cos\beta + 4\sin\beta\cos\beta) + \ddot{\zeta}$$

$$= \ddot{\zeta} + \left(\ddot{\zeta} - \zeta\dot{\beta}^2\right)\left(1 + 3\sin^2\beta\right) + 3\left(2\dot{\zeta}\dot{\beta} + \zeta\ddot{\beta}\right)\sin\beta\cos\beta$$

Right-hand side, term by term: $-mg\,\mathbf{k}_1 \cdot \mathbf{e}_\zeta = -mg$; $\mathbf{F}_C \cdot \mathbf{e}_\zeta = m\Omega^2\zeta\left(\cos^2\beta + 4\sin^2\beta\right) = m\Omega^2\zeta\left(1 + 3\sin^2\beta\right)$;

$$\mathbf{F}_{COR} \cdot \mathbf{e}_\zeta = 4m\Omega\left(\dot{\zeta}\sin\beta + \zeta\dot{\beta}\cos\beta\right)\cos\beta - 4m\Omega\left(\dot{\zeta}\cos\beta - \zeta\dot{\beta}\sin\beta\right)\sin\beta = 4m\Omega\zeta\dot{\beta}\left(\cos^2\beta + \sin^2\beta\right) = 4m\Omega\zeta\dot{\beta}$$

Dividing by m:

$$\boxed{\ddot{\zeta}+\left(\ddot{\zeta}-\zeta\dot{\beta}^2\right)\left(1+3\sin^2\beta\right)+3\left(2\dot{\zeta}\dot{\beta}+\zeta\ddot{\beta}\right)\cos\beta\sin\beta=-g+\Omega^2\zeta\left(1+3\sin^2\beta\right)+4\Omega\zeta\dot{\beta}}$$

Projection on $\mathbf{e}_\beta=(-\sin\beta,2\cos\beta,0)$. Left-hand side:

$$\mathbf{a}_1^P\cdot\mathbf{e}_\beta=\left(\ddot{\zeta}-\zeta\dot{\beta}^2\right)(-\sin\beta\cos\beta+4\sin\beta\cos\beta)+\left(2\dot{\zeta}\dot{\beta}+\zeta\ddot{\beta}\right)\left(\sin^2\beta+4\cos^2\beta\right)=3\left(\ddot{\zeta}-\zeta\dot{\beta}^2\right)\sin\beta\cos\beta+\left(2\dot{\zeta}\dot{\beta}+\zeta\ddot{\beta}\right)\left(1+3\cos^2\beta\right)$$

Right-hand side: $-mg\,\mathbf{k}_1\cdot\mathbf{e}_\beta=0$; $\mathbf{F}_C\cdot\mathbf{e}_\beta=m\Omega^2\zeta\left(-\cos\beta\sin\beta+4\sin\beta\cos\beta\right)=3m\Omega^2\zeta\sin\beta\cos\beta$;

$$\mathbf{F}_{COR}\cdot\mathbf{e}_\beta=-4m\Omega\left(\dot{\zeta}\sin\beta+\zeta\dot{\beta}\cos\beta\right)\sin\beta-4m\Omega\left(\dot{\zeta}\cos\beta-\zeta\dot{\beta}\sin\beta\right)\cos\beta=-4m\Omega\dot{\zeta}$$

Dividing by m:

$$\boxed{3\left(\ddot{\zeta}-\zeta\dot{\beta}^2\right)\cos\beta\sin\beta+\left(2\dot{\zeta}\dot{\beta}+\zeta\ddot{\beta}\right)\left(1+3\cos^2\beta\right)=3\Omega^2\zeta\cos\beta\sin\beta-4\Omega\dot{\zeta}}$$

These two equations contain only ζ, β and their derivatives. The initial conditions follow from the data: $\mathbf{r}_{1,0}^P=2\mathbf{j}_1+\mathbf{k}_1$ gives $\zeta_0=1$, $\zeta_0\cos\beta_0=0$ and $2\zeta_0\sin\beta_0=2$, so $\beta_0=\pi/2$; $\mathbf{v}_{1,0}^P=\mathbf{i}_1$ gives $\dot{\zeta}_0=0$ ($\mathbf{k}_1$ component) and $\dot{\zeta}_0\cos\beta_0-\zeta_0\dot{\beta}_0\sin\beta_0=-\dot{\beta}_0=1$, so $\dot{\beta}_0=-1$ rad/s; the $\mathbf{j}_1$ component $2(\dot{\zeta}_0\sin\beta_0+\zeta_0\dot{\beta}_0\cos\beta_0)=0$ is consistent (the initial velocity is tangent to the cone).

Answers and Checks

(a) Degrees of freedom

$\mathrm{DOF}=2$, with generalized coordinates (ζ,β).

(b) Kinematics in S_1

$\mathbf{r}_1^P$, $\mathbf{v}_1^P$ and $\mathbf{a}_1^P$ as in the kinematics section, with $\mathbf{i}_1=\cos\Omega t\,\mathbf{i}_0+\sin\Omega t\,\mathbf{j}_0$, $\mathbf{j}_1=-\sin\Omega t\,\mathbf{i}_0+\cos\Omega t\,\mathbf{j}_0$, $\mathbf{k}_1=\mathbf{k}_0$.

(c) Forces

Weight $-mg\,\mathbf{k}_1$, normal reaction $N\,\mathbf{n}$, centrifugal force $m\Omega^2\zeta(\cos\beta\,\mathbf{i}_1 + 2\sin\beta\,\mathbf{j}_1)$ and Coriolis force $2m\Omega(\dot{y}_1\,\mathbf{i}_1 - \dot{x}_1\,\mathbf{j}_1)$; the Euler force and the force of the origin's acceleration are zero.

(d) Equations of motion

The two boxed equations of the previous section, with $\zeta_0 = 1$ m, $\beta_0 = \pi/2$, $\dot{\zeta}_0 = 0$, $\dot{\beta}_0 = -1$ rad/s.

(e) Conservation law: mechanical energy in S_1

> **Pedagogical Justification:** E_1^P is conserved when every force that does work in S_1, inertia forces included, derives from a potential (Slide 29). Here: $\mathbf{N}\cdot\mathbf{v}_1^P = 0$ because the cone is fixed in S_1 and $\mathbf{v}_1^P$ is tangent to it; $\mathbf{F}_{COR}\cdot\mathbf{v}_1^P = 0$ always (Slide 31); $\mathbf{F}_A = \mathbf{F}_E = \mathbf{0}$; the weight is a constant force in S_1 ($\mathbf{k}_1 = \mathbf{k}_0$) with potential mgz_1; and, since $\boldsymbol{\omega}_{10}$ is constant, the centrifugal force has potential $-\tfrac{1}{2}m\Omega^2 r_\perp^2$ (Slide 30).

Kinetic energy in S_1, expanding the squares of the components of $\mathbf{v}_1^P$:

$$\left|\mathbf{v}_1^P\right|^2 = \left(\dot{\zeta}^2\cos^2\beta - 2\zeta\dot{\zeta}\dot{\beta}\sin\beta\cos\beta + \zeta^2\dot{\beta}^2\sin^2\beta\right) + 4\left(\dot{\zeta}^2\sin^2\beta + 2\zeta\dot{\zeta}\dot{\beta}\sin\beta\cos\beta + \zeta^2\dot{\beta}^2\cos^2\beta\right) + \dot{\zeta}^2$$

$$T_1^P = \frac{m}{2}\left[\left(2 + 3\sin^2\beta\right)\dot{\zeta}^2 + \left(1 + 3\cos^2\beta\right)\zeta^2\dot{\beta}^2 + 6\zeta\dot{\zeta}\dot{\beta}\sin\beta\cos\beta\right]$$

Potential in S_1, with $z_1 = \zeta$ and $r_\perp^2 = x_1^2 + y_1^2 = \zeta^2\left(\cos^2\beta + 4\sin^2\beta\right) = \zeta^2\left(1 + 3\sin^2\beta\right)$:

$$V_1 = mg\zeta - \frac{1}{2}m\Omega^2\zeta^2\left(1 + 3\sin^2\beta\right)$$

Direct check that $E_1^P = T_1^P + V_1$ is constant. Since $\mathbf{v}_1^P = \dot{\zeta}\,\mathbf{e}_\zeta + \zeta\dot{\beta}\,\mathbf{e}_\beta$ and $\dfrac{dV_1}{dt} = -\left(\mathbf{W} + \mathbf{F}_C\right)\cdot\mathbf{v}_1^P$ (chain rule on $V_1(\mathbf{r}_1^P(t))$):

$$\frac{dE_1^P}{dt} = \left(m\mathbf{a}_1^P - \mathbf{W} - \mathbf{F}_C\right)\cdot\mathbf{v}_1^P = \dot{\zeta}\left(m\mathbf{a}_1^P - \mathbf{W} - \mathbf{F}_C\right)\cdot\mathbf{e}_\zeta + \zeta\dot{\beta}\left(m\mathbf{a}_1^P - \mathbf{W} - \mathbf{F}_C\right)\cdot\mathbf{e}_\beta$$

and by the two equations of part (d) the brackets equal $4m\Omega\zeta\dot{\beta}$ and $-4m\Omega\dot{\zeta}$, so

$$\frac{dE_1^P}{dt} = \dot{\zeta}\left(4m\Omega\zeta\dot{\beta}\right) + \zeta\dot{\beta}\left(-4m\Omega\dot{\zeta}\right) = 0$$

Conservation law:

$$\boxed{\frac{m}{2}\left[\left(2+3\sin^2\beta\right)\dot{\zeta}^2 + \left(1+3\cos^2\beta\right)\zeta^2\dot{\beta}^2 + 6\zeta\dot{\zeta}\dot{\beta}\sin\beta\cos\beta\right] + mg\zeta - \frac{m}{2}\Omega^2\zeta^2\left(1+3\sin^2\beta\right) = E_1^P = m\left(\tfrac{1}{2}v_0^2 + g\zeta_0 - 2\Omega^2\zeta_0^2\right)}$$

with $v_0 = |\mathbf{v}_{1,0}^P| = 1$ m/s and $\zeta_0 = 1$ m, so numerically $E_1^P = m\left(\frac{1}{2} + g - 2\Omega^2\right)$ in SI units.

The constant comes from $t = 0$: $\zeta_0 = 1$ m, $\beta_0 = \pi/2$, $\dot{\zeta}_0 = 0$, $\dot{\beta}_0 = -1$ rad/s give $T_1^P(0) = \frac{m}{2}\,(1+0)\,(1)^2(-1)^2 = \frac{m}{2}$ (that is, $\frac{1}{2}m|\mathbf{v}_{1,0}^P|^2$ with $|\mathbf{v}_{1,0}^P| = 1$ m/s) and $V_1(0) = mg(1) - \frac{1}{2}m\Omega^2(1)^2(1+3) = m\left(g - 2\Omega^2\right)$, in joules when m, g and Ω are in SI units.

The angular momentum about Oz_0 is **not** conserved: the moment of $\mathbf{N}$ about the axis is $\left(\mathbf{r}_1^P \times \mathbf{n}\right)\cdot\mathbf{k}_1\,N = \left(x_1 n_y - y_1 n_x\right)N = \dfrac{\zeta\left(-\cos\beta\sin\beta + 4\sin\beta\cos\beta\right)}{\sqrt{5+3\cos^2\beta}}N = \dfrac{3\zeta\sin\beta\cos\beta}{\sqrt{5+3\cos^2\beta}}N$, which does not vanish in general because the section is not circular.

(f) Equilibrium positions in S_1

> **Pedagogical Justification:** an equilibrium in S_1 is a solution with ζ and β constant, so $\dot{\zeta} = \dot{\beta} = \ddot{\zeta} = \ddot{\beta} = 0$ in the two equations of part (d); the Coriolis terms vanish automatically.

The $\mathbf{e}_\beta$ equation reduces to $0 = 3\Omega^2\zeta\cos\beta\sin\beta = \frac{3}{2}\Omega^2\zeta\sin 2\beta$. With $\zeta > 0$ and $\Omega \neq 0$, $\sin 2\beta = 0$:

$$\beta \in \left\{0,\ \frac{\pi}{2},\ \pi,\ \frac{3\pi}{2}\right\}$$

The $\mathbf{e}_\zeta$ equation reduces to $0 = -g + \Omega^2\zeta\left(1+3\sin^2\beta\right)$, so

$$\zeta_{eq} = \frac{g}{\Omega^2\left(1+3\sin^2\beta\right)}$$

β	ζ_{eq}	**Position** $\mathbf{r}_1^P$
0	g/Ω^2	$\frac{g}{\Omega^2}(\mathbf{i}_1+\mathbf{k}_1)$
π	g/Ω^2	$\frac{g}{\Omega^2}(-\mathbf{i}_1+\mathbf{k}_1)$
$\pi/2$	$g/(4\Omega^2)$	$\frac{g}{2\Omega^2}\mathbf{j}_1+\frac{g}{4\Omega^2}\mathbf{k}_1$
$3\pi/2$	$g/(4\Omega^2)$	$-\frac{g}{2\Omega^2}\mathbf{j}_1+\frac{g}{4\Omega^2}\mathbf{k}_1$

Force-balance check at $\beta=0$: $\mathbf{n}=(-2,0,2)/\sqrt{8}$, so $\mathbf{0}=-mg\,\mathbf{k}_1+m\Omega^2\zeta\,\mathbf{i}_1+\frac{N}{\sqrt{2}}(-\mathbf{i}_1+\mathbf{k}_1)$ gives $N=\sqrt{2}\,mg$ and $\Omega^2\zeta=g$. **At $\beta=\pi/2$:** $\mathbf{n}=(0,-1,2)/\sqrt{5}$, so $\mathbf{0}=-mg\,\mathbf{k}_1+2m\Omega^2\zeta\,\mathbf{j}_1+\frac{N}{\sqrt{5}}(-\mathbf{j}_1+2\mathbf{k}_1)$ gives $N=\frac{\sqrt{5}}{2}mg$ and $2\Omega^2\zeta=\frac{g}{2}$, i.e. $\zeta=g/(4\Omega^2)$. Both agree with the table.

Extension (stability, not asked): the Hessian of $V_1(\zeta,\beta)$ is $\mathrm{diag}\left(-m\Omega^2,\ -3mg^2/\Omega^2\right)$ at $\beta=0,\pi$ and $\mathrm{diag}\left(-4m\Omega^2,\ 3mg^2/(16\Omega^2)\right)$ at $\beta=\pi/2,3\pi/2$. V_1 has no minimum (along each generator it is a downward parabola in ζ), so the energy criterion cannot certify the stability of any of these equilibria.

Checks

- **Meaning of the equilibria:** they lie on the four generators contained in the principal planes x_1z_1 and y_1z_1 of the cone (the sections drawn above). Only there does the normal $\mathbf{n}$ lie in the vertical plane through the axis, so that it can balance the weight and the (radial) centrifugal force without a tangential remainder. On the steeper generators ($\beta=0,\pi$, slope 45°) the particle must be higher, $\zeta=g/\Omega^2$, than on the flatter ones ($\beta=\pm\pi/2$, $\zeta=g/(4\Omega^2)$).
- **Gyroscopic coupling:** the Coriolis terms $4\Omega\zeta\dot\beta$ and $-4\Omega\dot\zeta$ couple the two equations antisymmetrically, which is why they cancel in dE_1^P/dt (Coriolis never does work).
- **Limit $\Omega\to0$:** the centrifugal and Coriolis terms vanish, E_1^P becomes the ordinary energy $\frac12mv^2+mgz$, and $\zeta_{eq}\to\infty$: a fixed, vertex-down smooth cone has no equilibrium position with $\zeta>0$. Basis: $t=0$ gives $[{}_0R_1]=I$, the two frames coincide as stated.
- **Orthogonality:** $\mathbf{n}\cdot\mathbf{e}_\zeta=\mathbf{n}\cdot\mathbf{e}_\beta=0$ (computed in part (d)); $\mathbf{e}_\zeta$ and $\mathbf{e}_\beta$ are not orthogonal in general, $\mathbf{e}_\zeta\cdot\mathbf{e}_\beta=3\sin\beta\cos\beta$, which is why the projections of part (d) mix the two accelerations.
- **Dimensions:** in the $\mathbf{e}_\zeta$ equation, $[\ddot\zeta]=[\zeta\dot\beta^2]=[\dot\zeta\dot\beta]=[\zeta\ddot\beta]=\mathrm{m/s^2}$, $[g]=\mathrm{m/s^2}$, $[\Omega^2\zeta]=\mathrm{s^{-2}\,m}$, $[\Omega\zeta\dot\beta]=\mathrm{s^{-1}\,m\,s^{-1}}$; the $\mathbf{e}_\beta$ equation has the same terms. $[E_1^P]=\mathrm{kg\,m^2/s^2=J}$; $[g/\Omega^2]=(\mathrm{m/s^2})/\mathrm{s^{-2}}=\mathrm{m}$.
- **Symbolic check (sympy):** $\mathbf{a}_0^P$ obtained by differentiating $[{}_0R_1]\,\mathbf{r}_1^P$ twice equals $\mathbf{a}_1^P+\boldsymbol{\omega}_{10}\times(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^P)+2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^P$ (Notes Eq. 3.27); both projections match the key; $dE_1^P/dt=0$ along the equations of motion; the four equilibria satisfy both equations.

- **Check against the official key:** both equations of part (d) coincide with the official key (Problems.pdf, p. 75) term by term.

Source: Problems.pdf (PDF page 25); key on PDF page 75